Langning Li

Phone: +1 (585) 230-1314 | Email: lli81@u.rochester.edu | GitHub: github.com/ton198 | Homepage: ton198.github.io/en.html

Class of 2028 | Available for Summer 2027 internships | Full-time on site, 5 days a week

Education

University of Rochester Expected May 2028

B.S. in Computer Science and B.A. in Mathematics (double degree) **GPA: 3.9/4.0**

Coursework: Machine Learning Theory, Deep Learning, Algorithm Design and Optimization, High-Performance Computing, Programming Language Design, Computer Organization

Technical Skills

- **Programming languages:** Python, TypeScript/JavaScript, Java, C, SQL.
- **Machine learning:** SFT, LoRA, SVD, PyTorch, Hugging Face Transformers, NumPy, Slurm GPU clusters.
- **Engineering:** Vue 3, Node.js, Docker, Playwright, Nginx, Git, Linux; Agent Skills (Claude Code / Codex / Oh My Pi).

Research and Internship Experience

University of Rochester | Research Assistant | Advisor: Xiangxiang Xu Sep 2026 – Present

LM-SVD: Analyzing the Low-Rank Structure of Language Model Outputs with NeuralSVD github.com/ton198/lm-svd

- Background: An ICLR 2026 paper showed that a language model's logits across contexts are highly correlated and can be approximated by a few directions, verified via SVD of the logits matrix. This project uses NeuralSVD (ICML 2024), which learns a matrix's leading singular directions with neural networks, to study this structure at scales classical SVD cannot handle.
- Applied NeuralSVD to full-vocabulary logits matrices, training on sampled contexts to approximate the model's output distribution; designed and trained a long-text embedding model to represent the singular functions so they generalize across contexts.
- Building on NeuralSVD, improved the original paper's SVD-based approximation of the output distribution (LinGen): the new approach skips solving for the original distribution with a small model, so generation no longer needs conventional LLM inference, greatly cutting LinGen's time and cost while lowering KL divergence by 36.8%.

Chinese Academy of Sciences | Research Assistant | Advisor: Zhengxin Yang May 2026 – Sep 2026

Younger: Crawling and Cleaning Model Computation Graph Data github.com/Yangs-AI/Younger-Logics-IR

- Background: For models trained by gradient descent, architecture largely determines performance; analyzing a model's computation graph can partly predict its performance and guide architecture optimization for different tasks. The project crawls as many computation graphs as possible from Hugging Face to build a dataset for training GNNs that analyze and optimize them.
- Developed computation graph extractors for models of different formats and versions, unifying them into the ONNX intermediate representation and storing the extracted graphs as DAGs for model training.
- Built an LLM-based extractor that, with a constrained output format, pulls tasks, datasets, and evaluation metrics from non-standardized Hugging Face model READMEs to build an evaluation dataset.

Auto Growth | Software Engineering Intern May 2026 – Sep 2026

AI Interactive Picture Book Production Platform and Agent Automation Workflows

- Background: The company supplies AI interactive picture books in bulk to Epic!, a U.S. digital reading platform for children. Each book includes pages, audio and video, character dialogue, and embedded mini-games; manual assembly, page-by-page preview, and asset checks made the process slow and error-prone. We set out to build an editing and production platform and automate every step from proposal to delivery with Agent workflows, enabling production at scale.
- Developed the team's internal interactive book editor and reader (Vue 3 + Node.js). Built a framework for interactive controls with development and debugging tools that simulate the reader runtime, load controls outside the editor, tune parameters in real time, and test interactions with external hooks. This greatly sped up interactive content development; nearly 500 standalone games have been completed.
- Packaged repetitive work such as book assembly, game porting, and asset checks into Agent Skills and workflows that automate the full pipeline from a proposal: asset generation and checking, book assembly, game generation and adaptation, and content review. This cut the production cycle per book by 90% and substantially reduced costs.
- Built a compatibility layer that adapts and delivers each book to Epic! automatically; delivered about 500 interactive picture books in the first batch.

Campus Activities and Honors

- **Tutor, CS Undergrads (CSUG) | Jan 2026 – Present:** Organize CS outreach for the club; tutor CS undergraduates in their courses.
- **CS Workshop Leader | Jan 2025 – Present:** Lead weekly 75-minute CS workshops: problem walkthroughs, Q&A, exam review.
- **HiMCM Finalist | Dec 2022:** Second prize; used an LSTM to forecast bee populations across U.S. states.